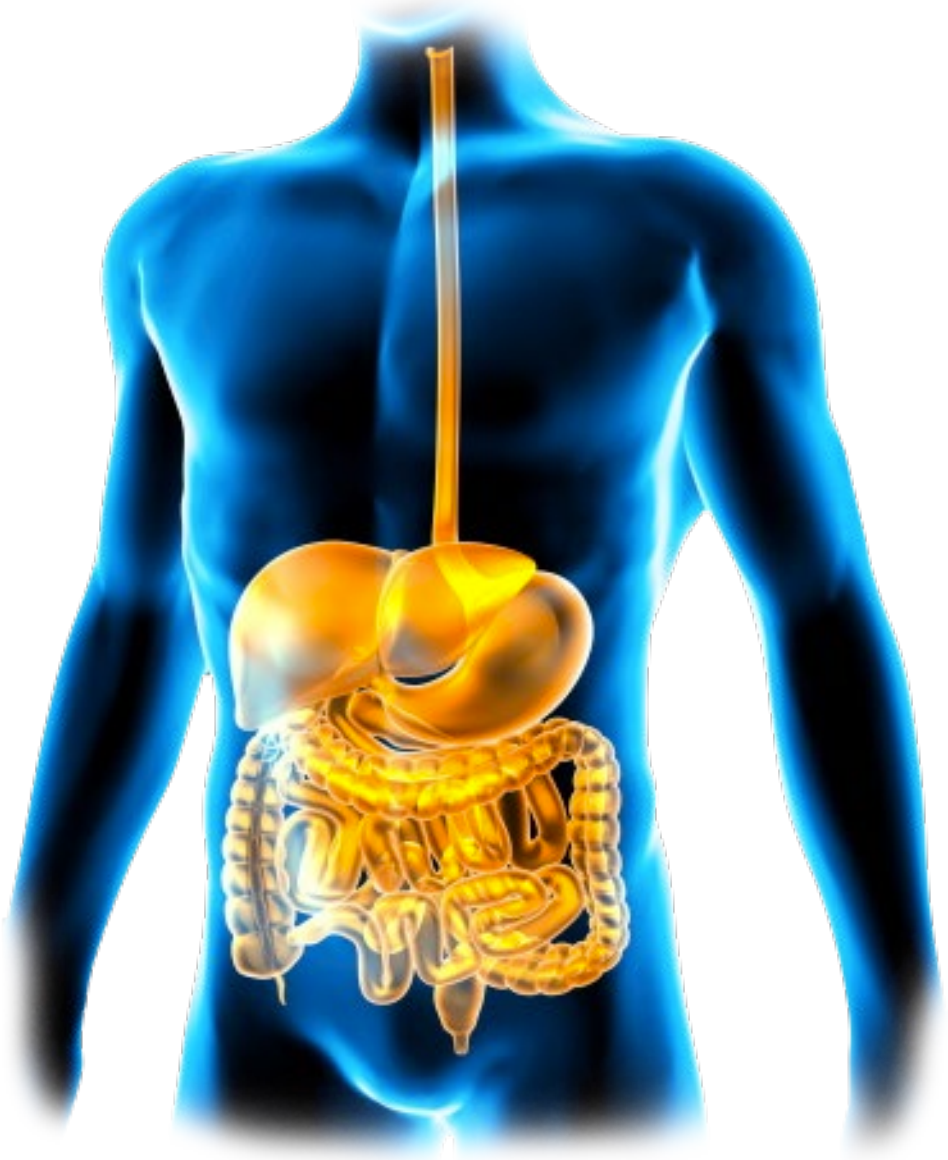
Visualizing the GI system

Fordøjelsessytem, patofysiologi og diagnostik

Pablo Pennisi, Associate Prof.
Regenerative Medicine group
2026



AALBORG UNIVERSITY
DENMARK

Læringsmål

- *Beskrive metoder til visualisering af fordøjelsessystemet*
- *Gøre rede for indikation af billeddiagnostiske undersøgelser og hvordan patientens symptomer påvirker undersøgelsesforløb*

Lecture overview

- Outline of the different methods
  - Endoscopic
  - Ultrasound-based
  - X-ray-based devices
  - MRI
- Criteria for method selection and examples

ENDOSCOPIC TECHNIQUES



Endoscopy

Advantages

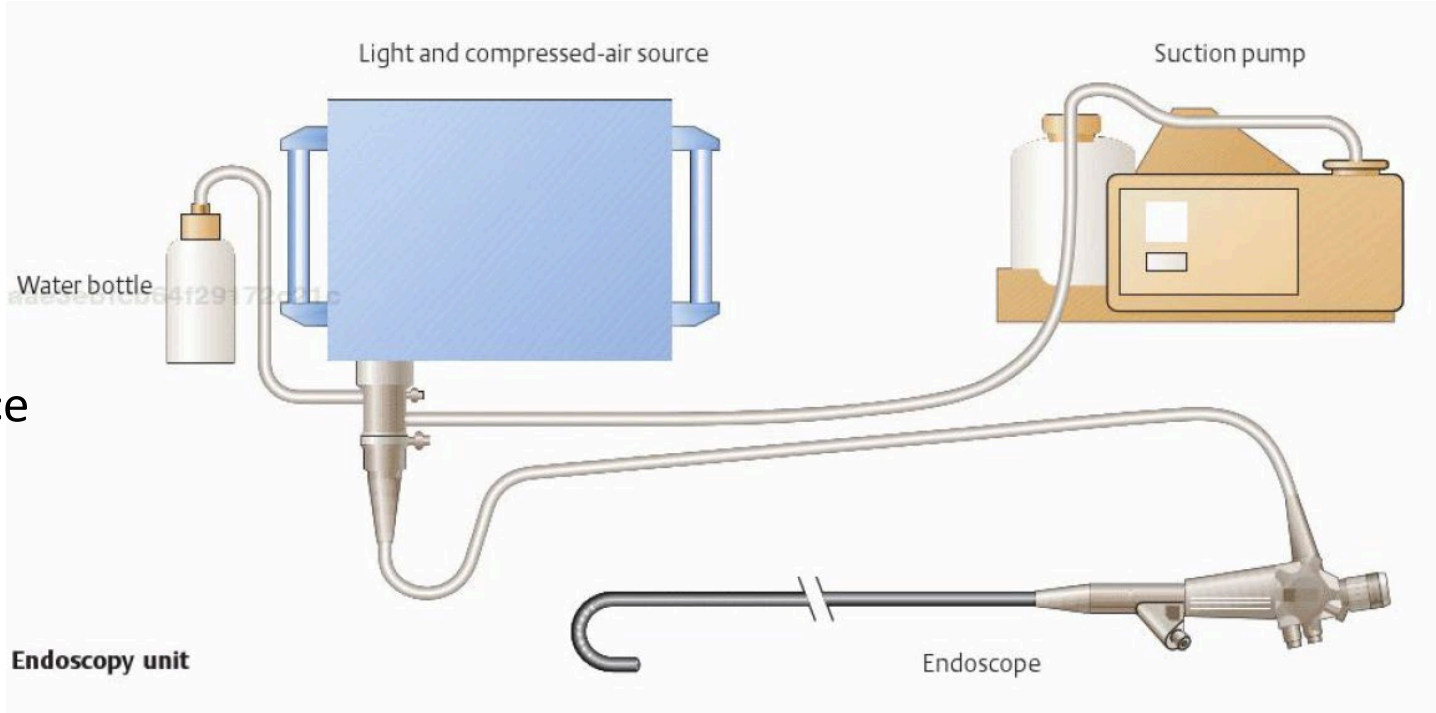
- Direct visualization of mucosal surface
- Allows taking biopsies
- Allows therapeutic procedures

Contraindications

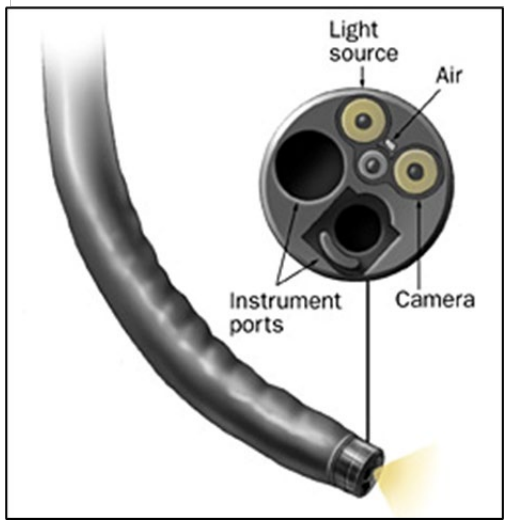
- Patients sensitive to sedation
- Uncooperative patients
- Patients with severe bleeding
- Diverticular disease

Limitations

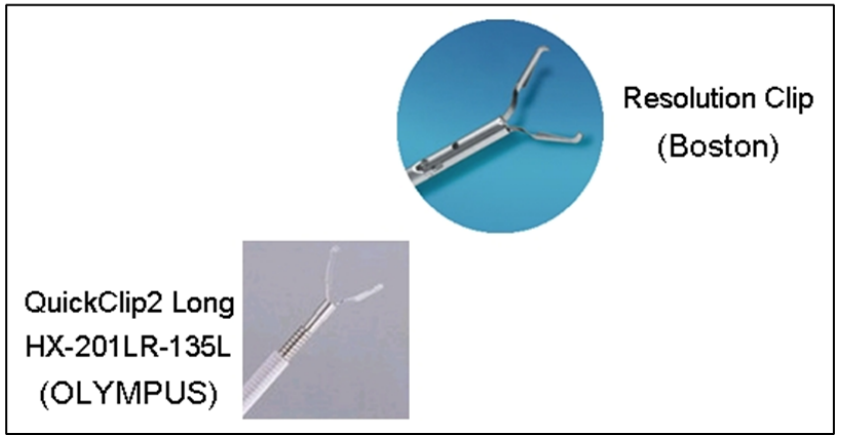
- Relatively expensive instrument
- Poor for motility measurements
- Poor for anatomy



Detail of the endoscope tip



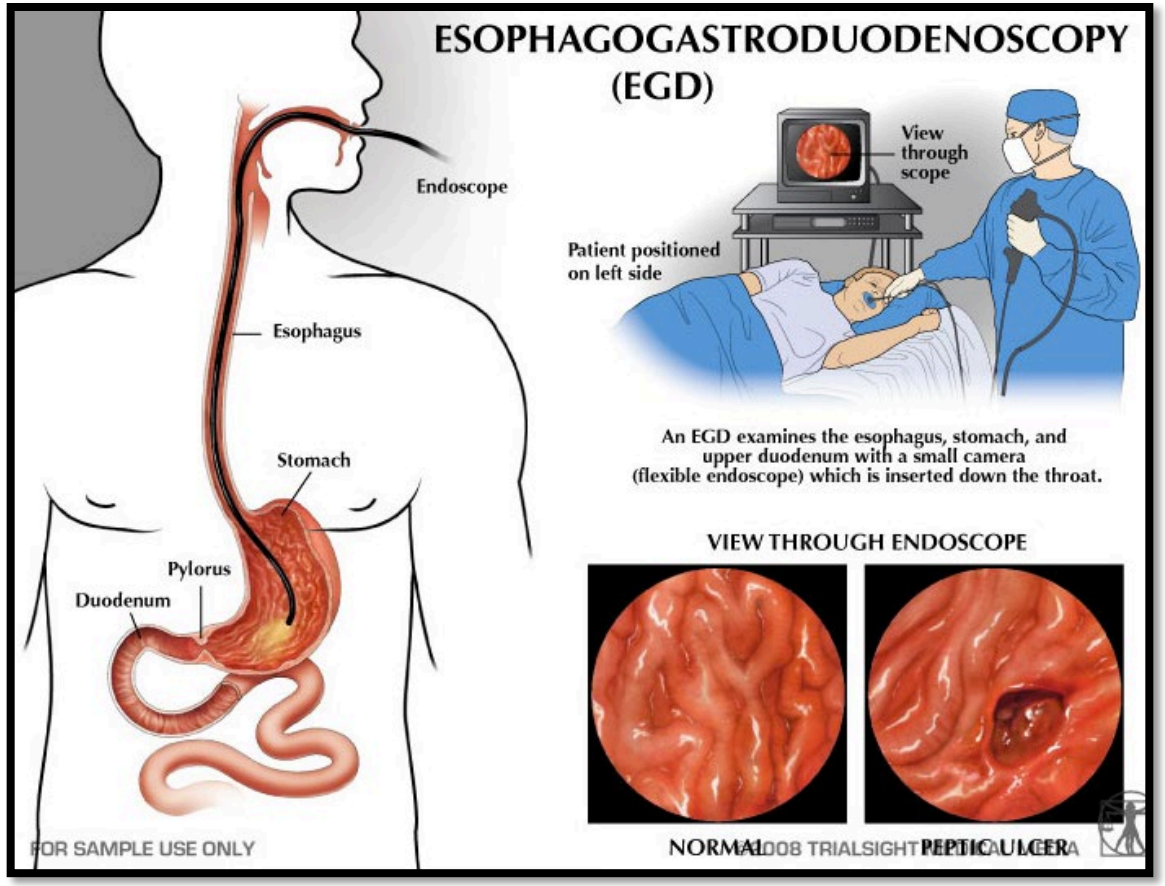
Hemostasis clips



Upper endoscopy

Indications

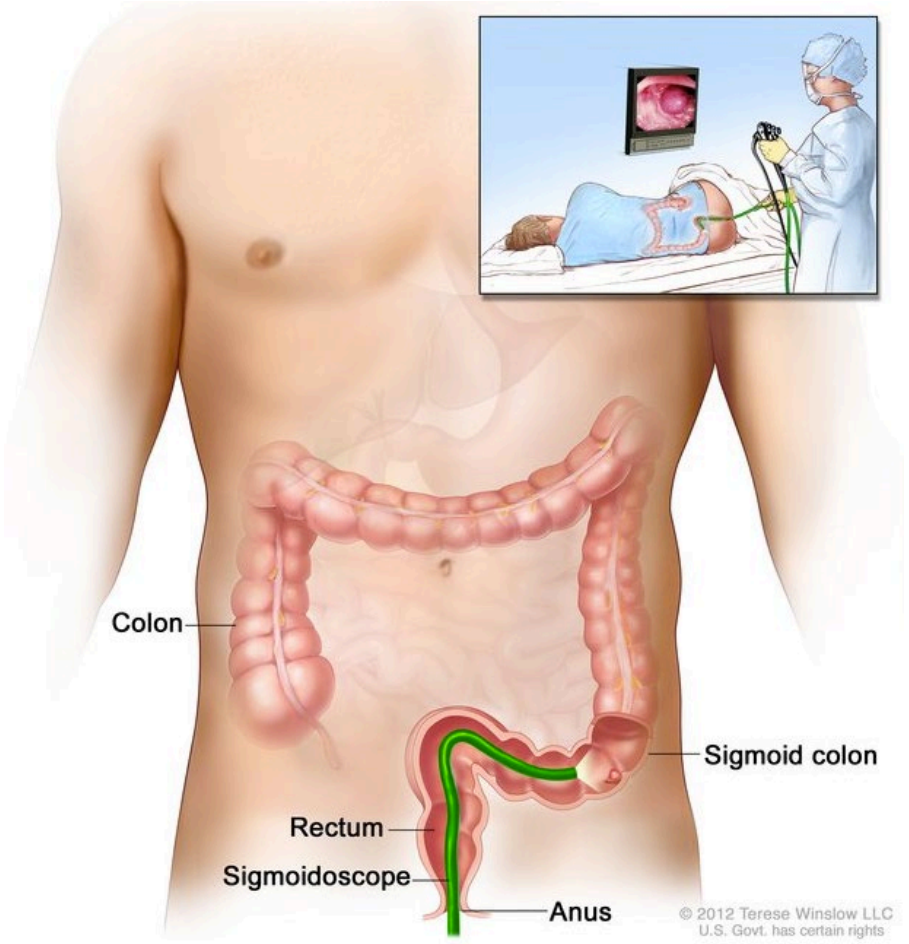
- Dyspepsia or suspicion of stomach cancer
- Investigation of:
  - Upper GI bleeding
  - Dysphagia
  - Gastroesophageal reflux disease (GERD)
  - Upper GI ulcers
- Surveillance of:
  - Healing lesions
  - Neoplasms



Lower endoscopy: Rectosigmoideoscopy

Indications

- Suspicion of colorectal cancer in the rectum/sigmoideum
- Assessment/control of IBD (UC)
- Therapeutic procedures in the last 60 cm of colon



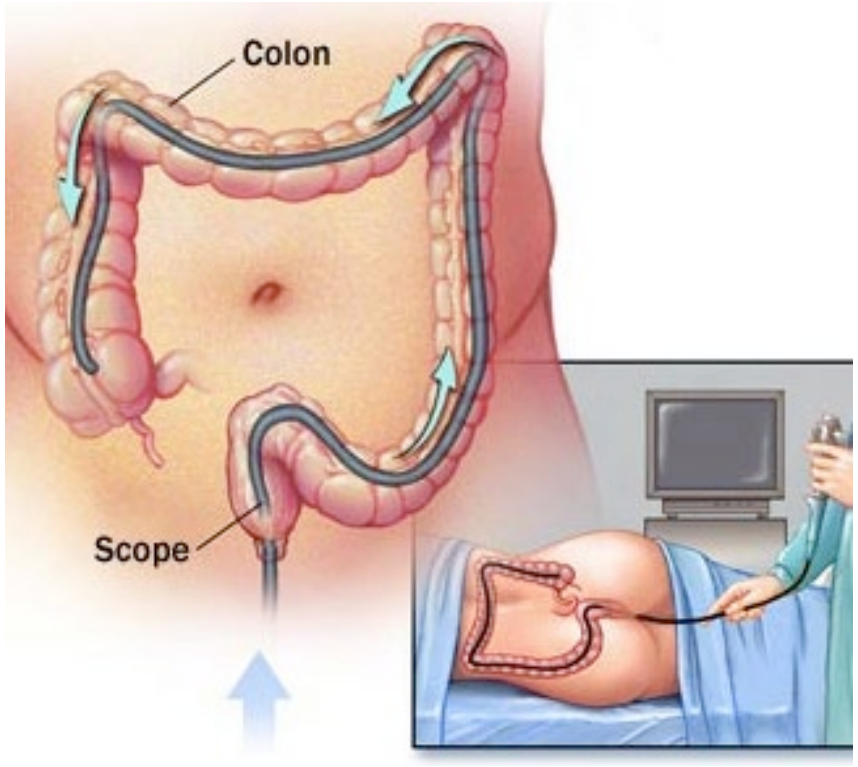
Lower endoscopy: Colonoscopy

Indications

- Examination of colon (up to the caecum and terminal ileum)
- Suspicion of colorectal cancer
- Assessment/control of IBD (Mb. Crohn)
- Therapeutic procedures

Limitations

- Risk of iatrogenic perforation



Capsule endoscopy

Alternative to endoscopic/radiologic examination of the small intestine

Indications

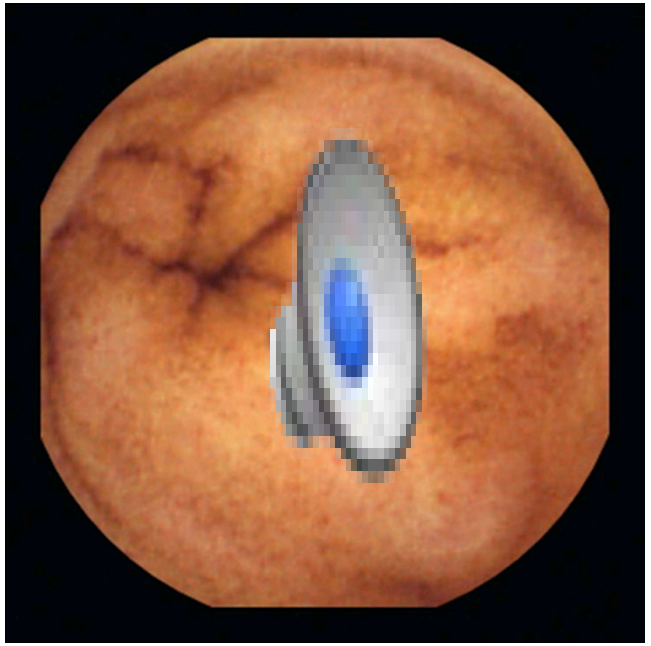
- Patients with GI bleeding, in which diagnosis by EGD or colonoscopy failed
- Assessment of Crohn's disease patients

Contraindications

- Patients with known or suspected GI obstructions
- Patient with active implanted devices
- Patients with swallowing disorders





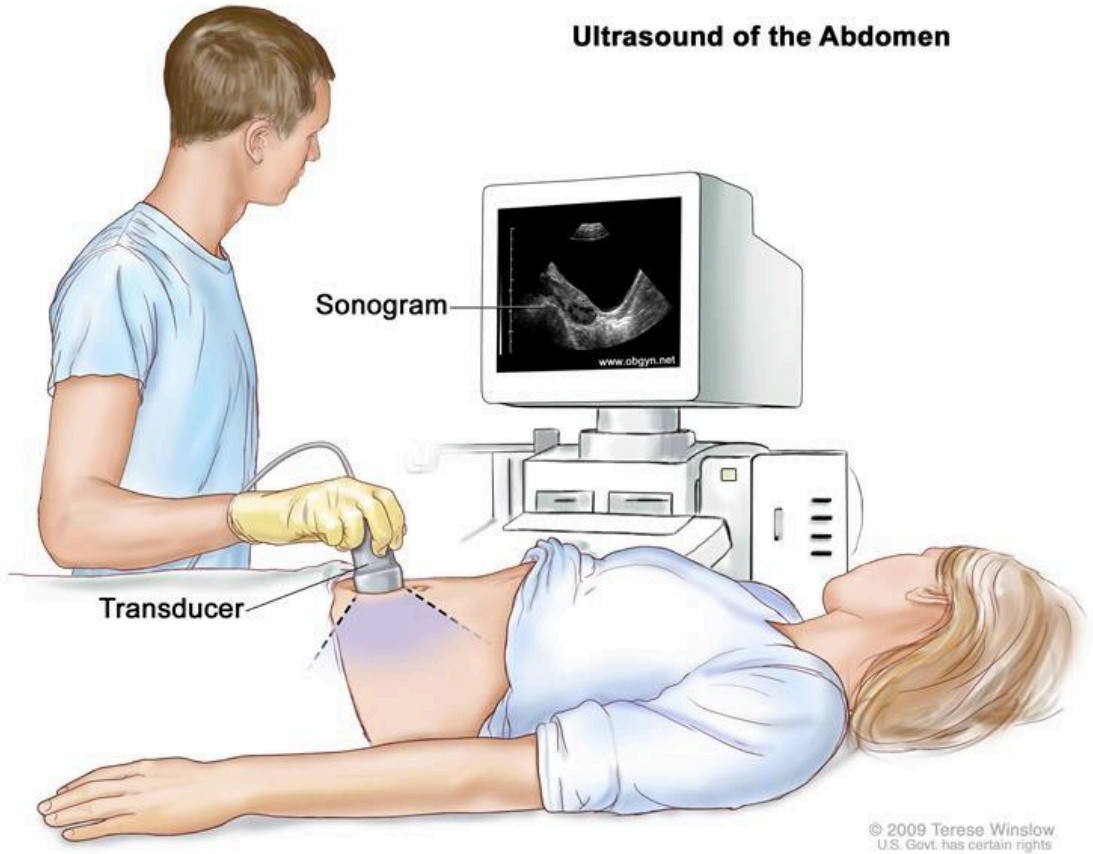


ULTRASOUND TECHNIQUES



Abdominal ultrasound

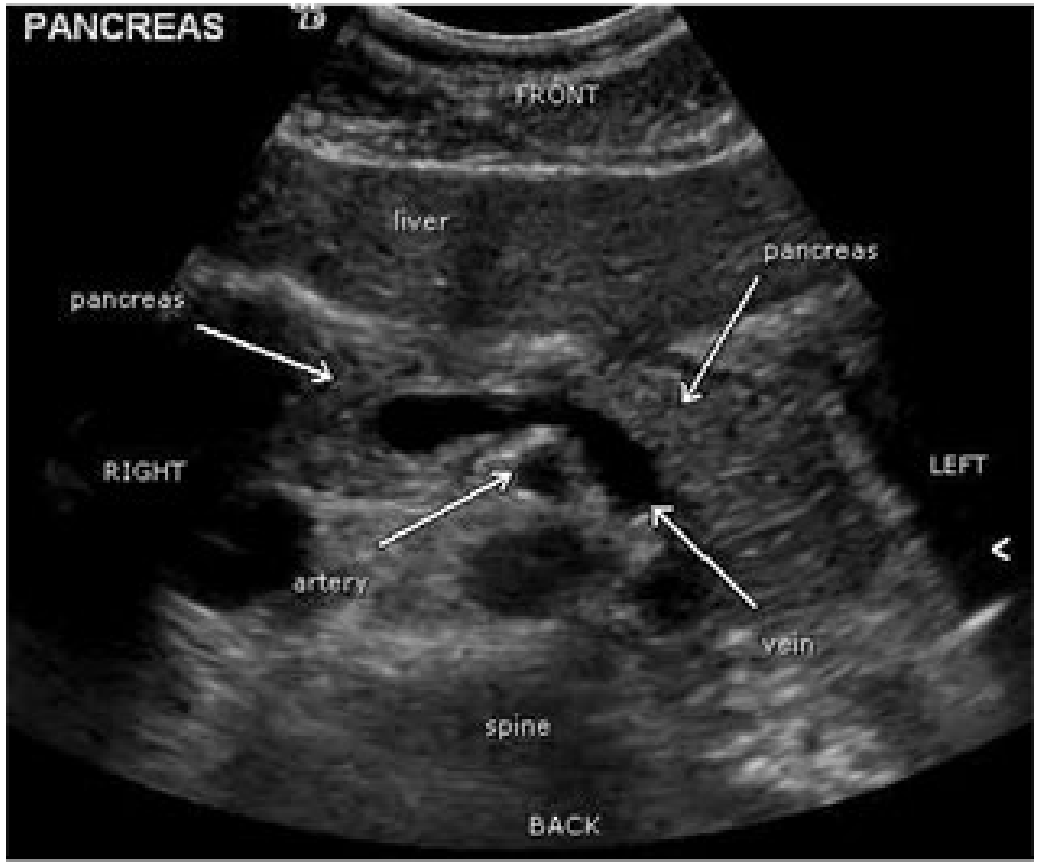
- Advantages
  - Cost effective
  - Adequate visceral visualization
  - Best for gallbladder
  - No radiation
- Limitations
  - Need a skilled operator
  - Poor
    - + In obese patients
    - + In presence of bowel gasses
    - + Below bone



Abdominal ultrasound

Indications

- Liver: masses, cysts, bile ducts, veins
- Biliary tract: gallstones, cholecystitis
- Pancreas: pancreatitis, cysts, tumours
- Others: Ascites, organomegalies, lymph nodes, intrabdominal masses



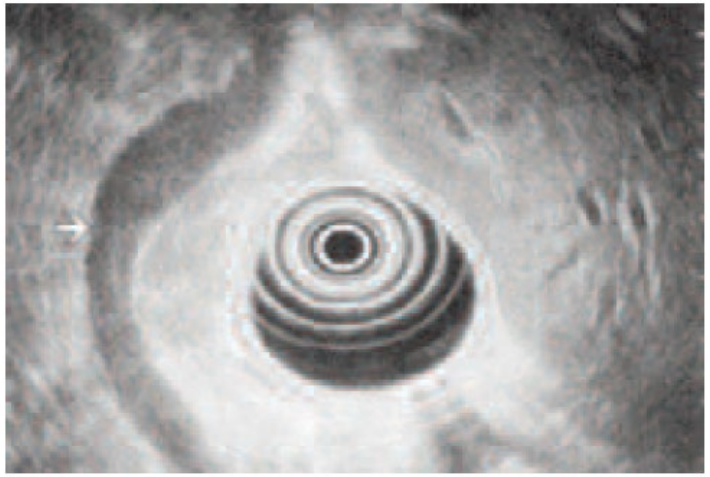
"need a skilled operator"

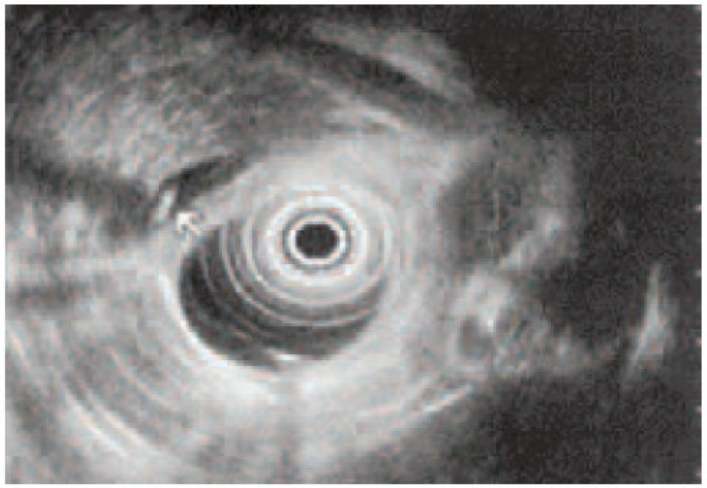
Endoscopy with ultrasound (EUS)

- Uses an ultrasound probe (echoendoscopy)
- Allows assessment of the GI wall and neighboring organs structure

Indications

- T and N assessment
- Visualization of stones in the common bile duct
- Assessment of submucosal changes throughout the GI





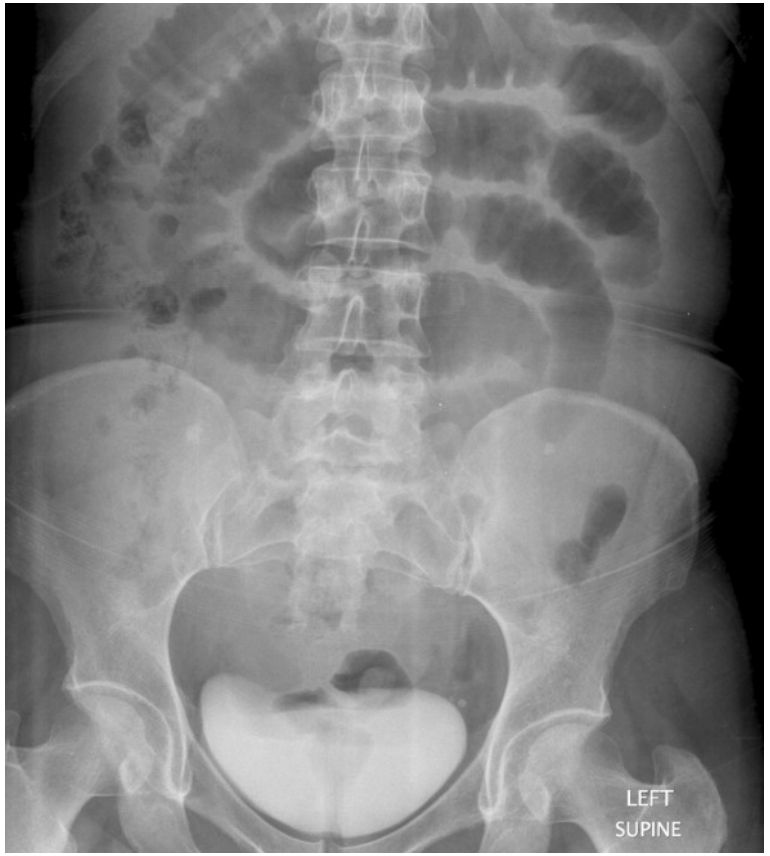
Ainsworth AP, et al.”Endoskopisk UL-skanning” (2003) Radiograffaglige artikler. Radiograf rådet https://www.radiograf.dk/radiografen/radiograffaglige-artikler/faglige-artikler-2003/

X-RAY BASED DEVICES



Plain x-ray films

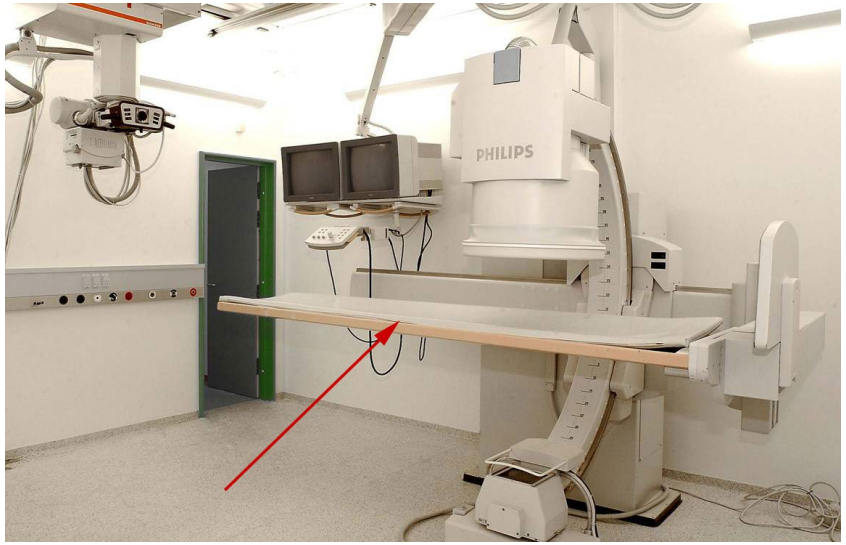
- X-rays allow visualization of different densities (air, fat, soft tissue, water, bone)
- Main advantage is low cost and wide availability
- Limited to few abdominal diseases: perforations, abnormal calcifications, bowel obstructions.



The plain film shows dilated loops of small bowel filled with gas.

Plain X-ray with contrast

To better visualize lumen and to differentiate healthy/diseased tissue



Limitations:

- Requires delivery of a contrast agent
- Requires patient cooperation



Contrasting agents

- Intraluminal
  - Non-water soluble
    - Barium sulfate
  - Water Soluble
    - Gastrografin
- Intraarterial/venous
  - Radioopaque: Iodine based



Contrast enemas: Barium

- Single Contrast
  - Generally uses just thin Barium (low viscosity)
  - Distends the lumen
  - Easier for patient, but less mucosal detail
- Double Contrast (air contrast)
  - Thick barium (high viscosity) coats lumen
  - Effervescent tablets ingested to distend lumen with air
  - Produces ‘see-through’ images with greater mucosal detail
  - Greater sensitivity for small lesions, polyps, ulcers

Single Contrast vs Double Contrast Barium enemas



Single Contrast
Barium Enema



Double Contrast
Barium Enema

Contrast enemas

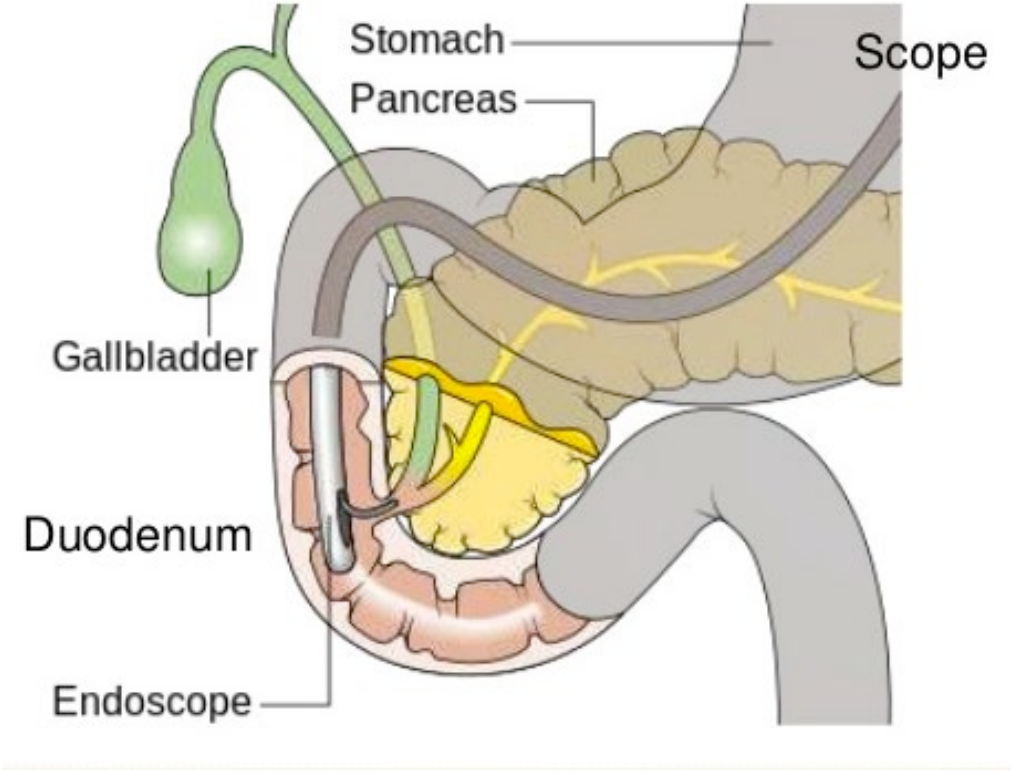
Indications

- **Single contrast:** Patient under 40 years of age with abdominal signs or symptoms not suggestive for polyps, colitis, or bleeding. In general for uncooperative, disabled, very old, or very ill patients
- **Double contrast:** Rectal bleeding, polyps or carcinoma, IBD. Patients over 40 years of age who can cooperate and turn over without assistance.

Contraindications to barium enema of any type

- Suspected acute perforation
- Complicated ulcerative colitis (toxic megacolon)
- Immediately after biopsy

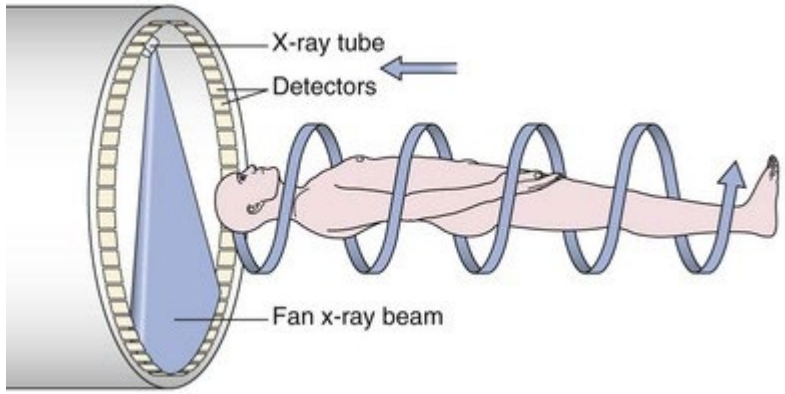
Endoscopic Retrograde Cholangiopancreatography (ERCP)

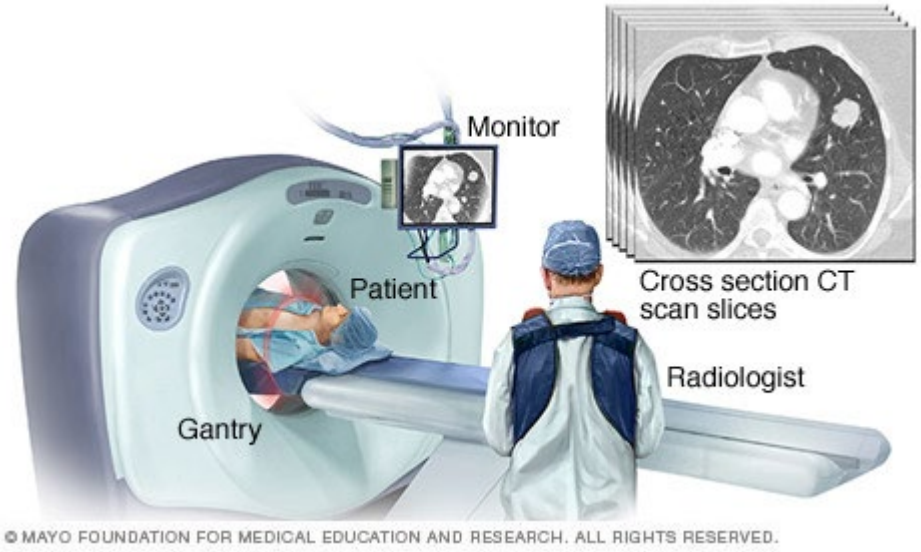




Computed tomography (CT)

- A standard tube produces high-energy X-rays
- A fan-shaped beam passes through the patient and is detected to form a projection.
- The source and detector are rotated to obtain a larger number of different projections.
- Images give a measure of the X-ray attenuation coefficient at every point.
- Limitation: exposure of the patient to ionizing radiation





When to use CT?

Best when we need:

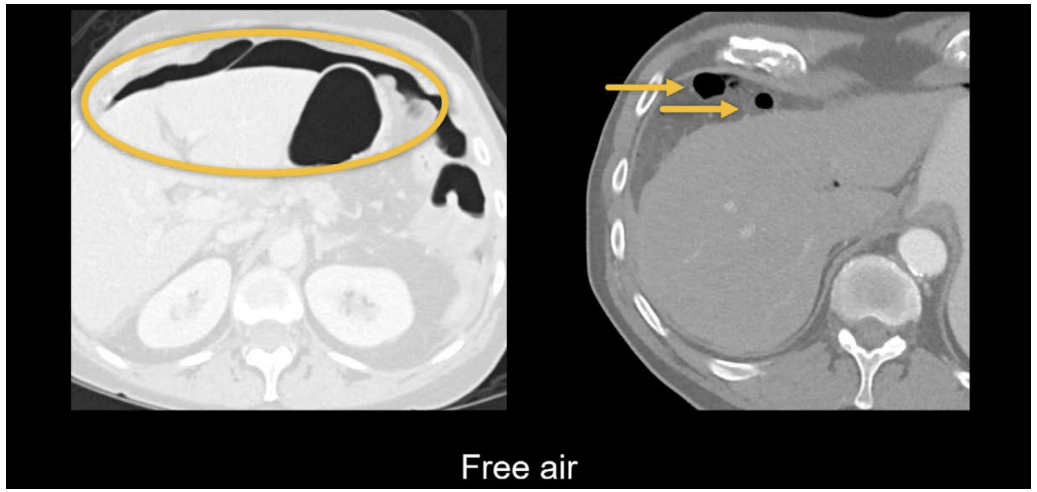
- A rapid overview of the entire abdomen
- Detection of inflammation
- Detection of perforation or obstruction
- Detection of abscesses and collections
- Cancer staging and detection of metastases

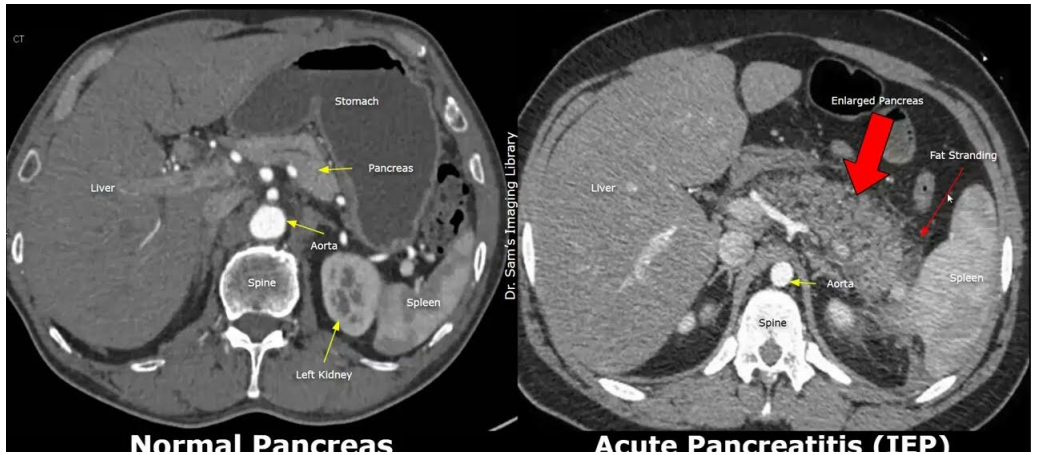
CT: typical indications

Acute abdomen
- Perforation
- Bowel obstruction
- Appendicitis
- Diverticulitis

Pancreas
- Acute pancreatitis
- Pancreatic tumors

Cancer
- Colon cancer staging
- Gastric cancer staging
- Detection of liver metastases



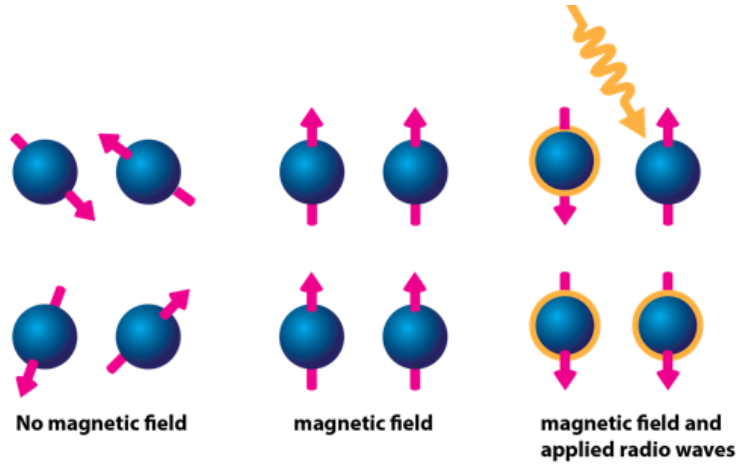


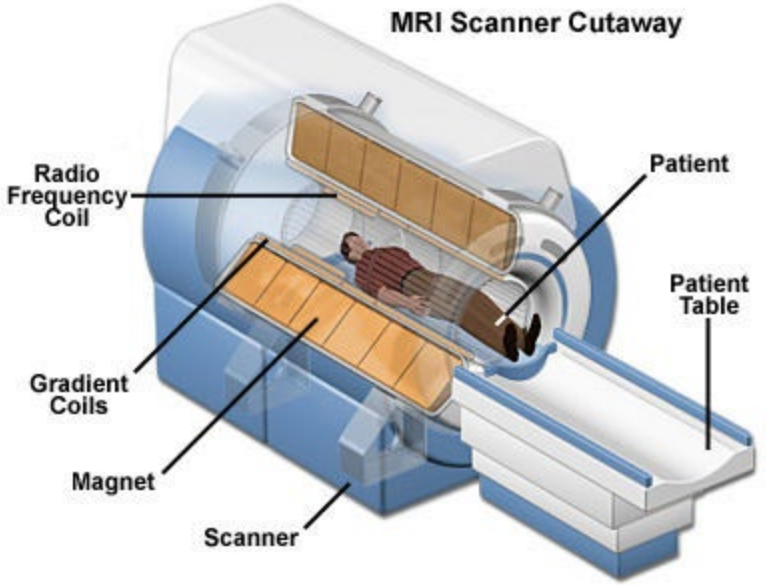
MAGNETIC RESONANCE IMAGING (MRI)



Magnetic resonance imaging (MRI)

- Based on NMR principles
- The human body is primarily in fat and water. It contains approx. 63% of H atoms
- MRI can distinguish tissues based on:
  - Proton density (PD)
  - Longitudinal relaxation time (T1)
  - Transverse relaxation time (T2)





Abdominal MRI

Indications

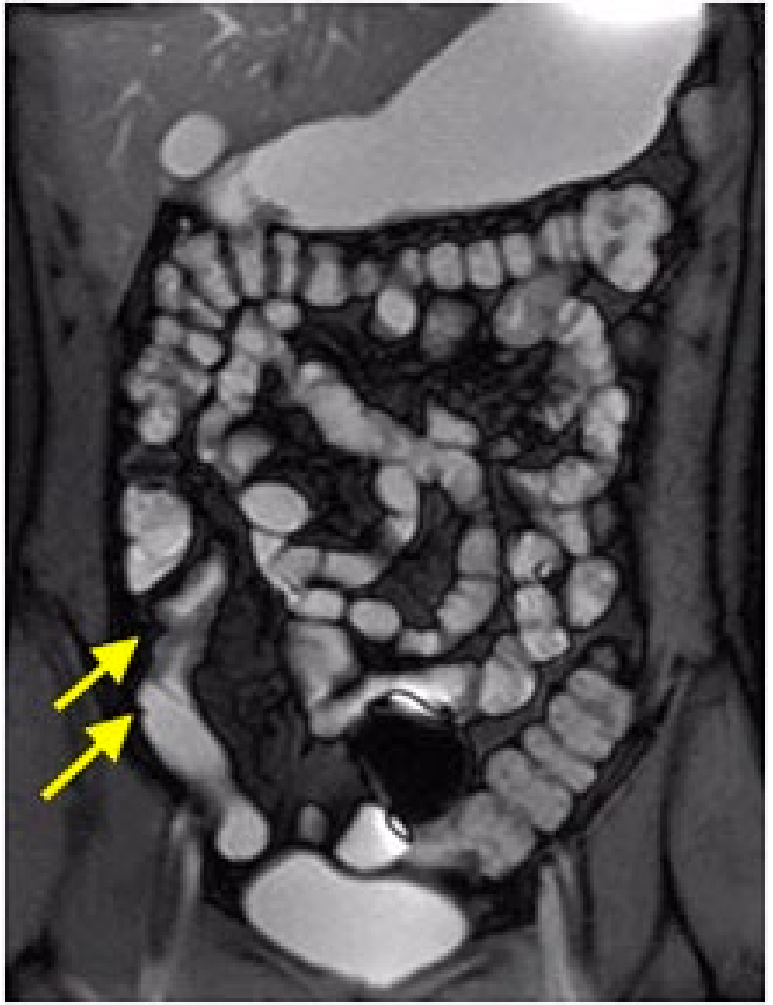
- Complementary to CT and superior for selected indications:
- Rectal cancer staging, liver lesions, MRCP (bile ducts), Crohn's disease (follow-up), perianal fistulas

Contraindications

- Patients bearing pacemakers

Limitations

- Cost
- Claustrophobic patients

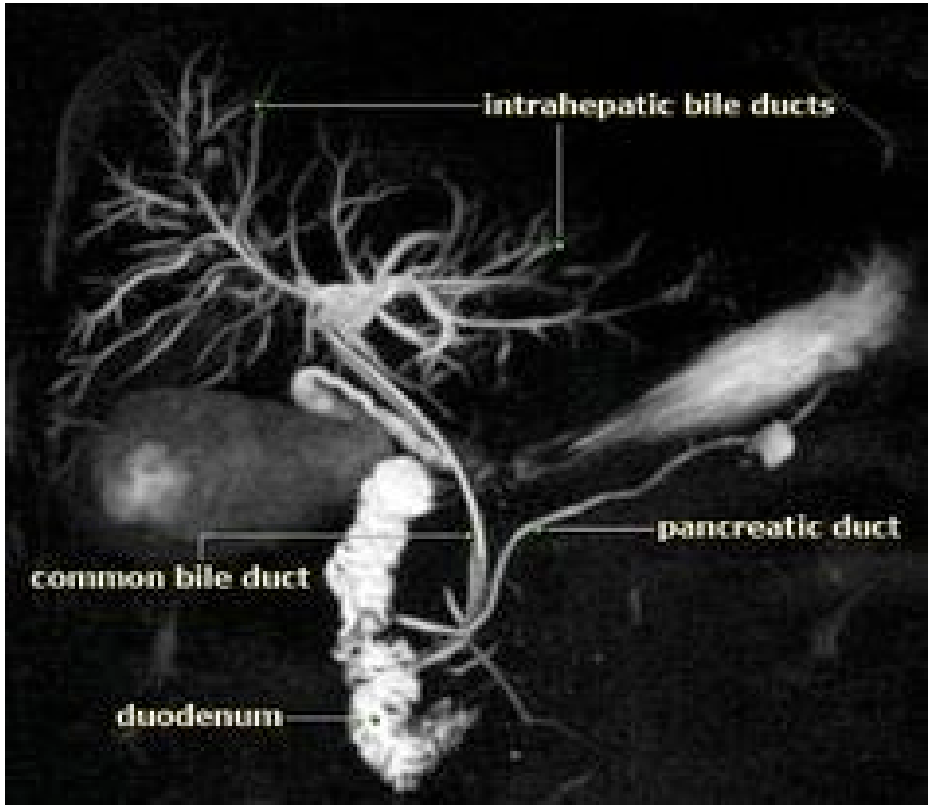


MR enterography shows mild Crohn’s disease involving the terminal ileum.

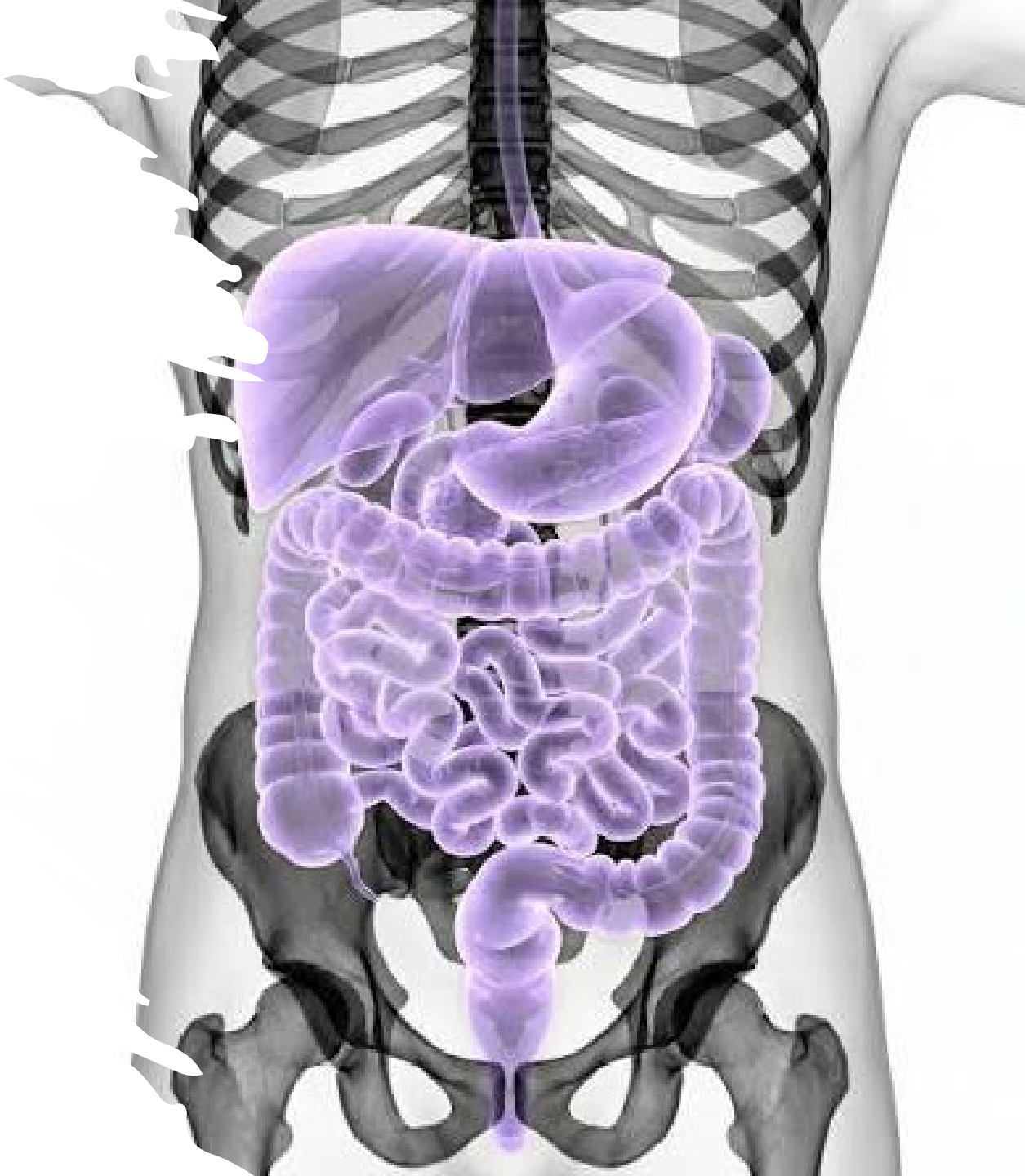
Abdominal MRI

Magnetic resonance cholangiopancreatography (MRCP)

- A rapid, accurate and noninvasive means to evaluate the gallbladder, bile ducts and pancreatic ducts.
- Equivalent accuracy to ERCP to detect stones, strictures and variant anatomy



Criteria for method selection and examples



Criteria for selection of imaging modality

- Which organ is most likely affected?
- Do we need to see the mucosa, the wall, the ducts, or the entire abdomen?
- Do we need a biopsy?
- What is the purpose: diagnosis, follow-up, or staging?

Example: acute abdominal pain

Recommended Imaging Studies Based on Location of Abdominal Pain

LOCATION OF PAIN	*IMAGING*
Right upper quadrant[14]	Ultrasonography
Left upper quadrant	CT
Right lower quadrant[15]	CT with IV contrast media
Left lower quadrant[16]	CT with oral and IV contrast media
Suprapubic	Ultrasonography



CT = computed tomography; IV = intravenous.

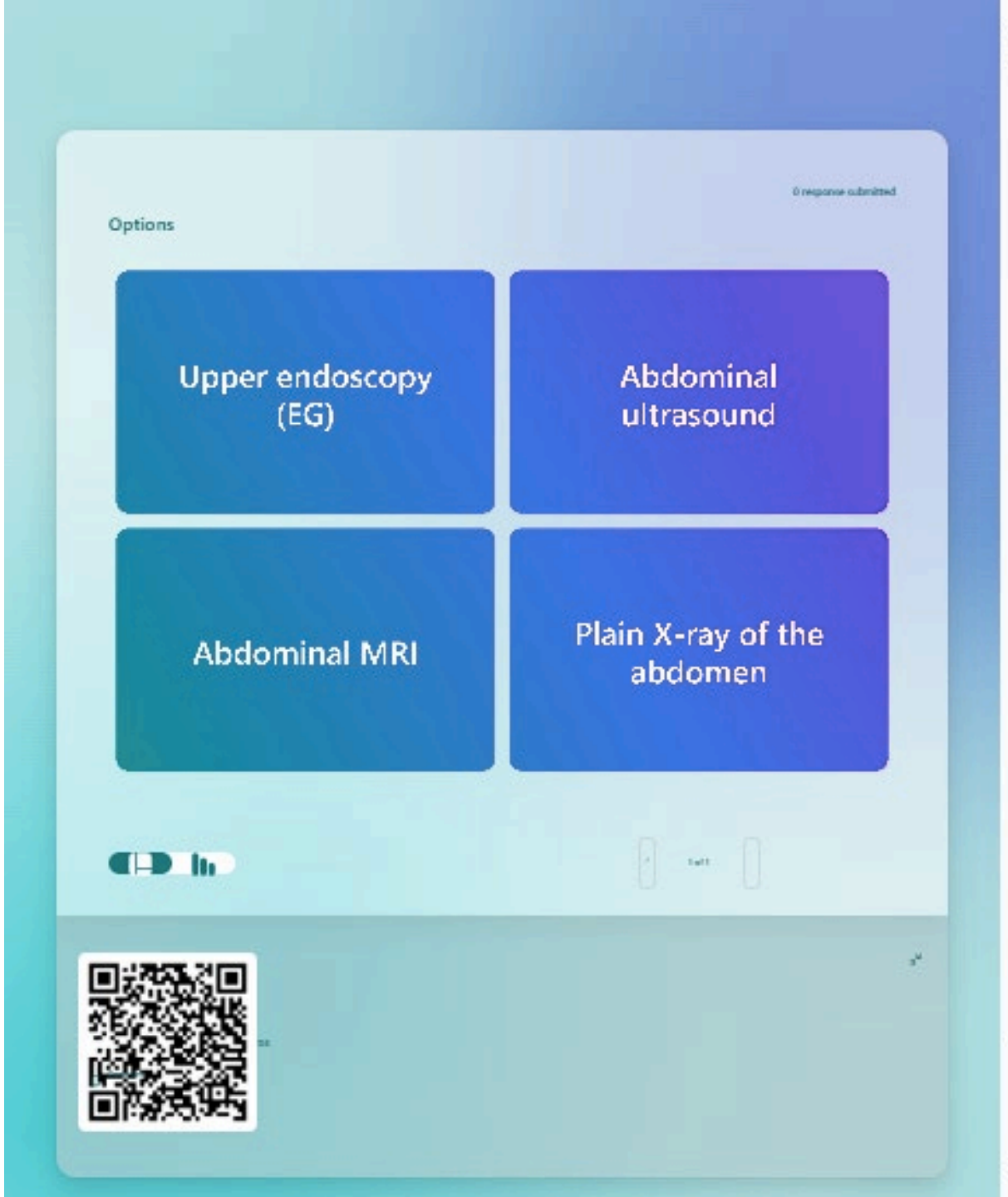
Cartwright SL, Knudson MP. Evaluation of acute abdominal pain in adults. Am Fam Physician 2008; 77: 971-8.

Case 1

A 58-year-old man presents with:

- Burning epigastric pain
- Symptoms for several months
- Pain improves after meals
- Dark stools during the last week
- Mild iron-deficiency anemia

Which imaging method would most likely provide the diagnosis?



Duodenal and gastric ulcers

Upper endoscopy

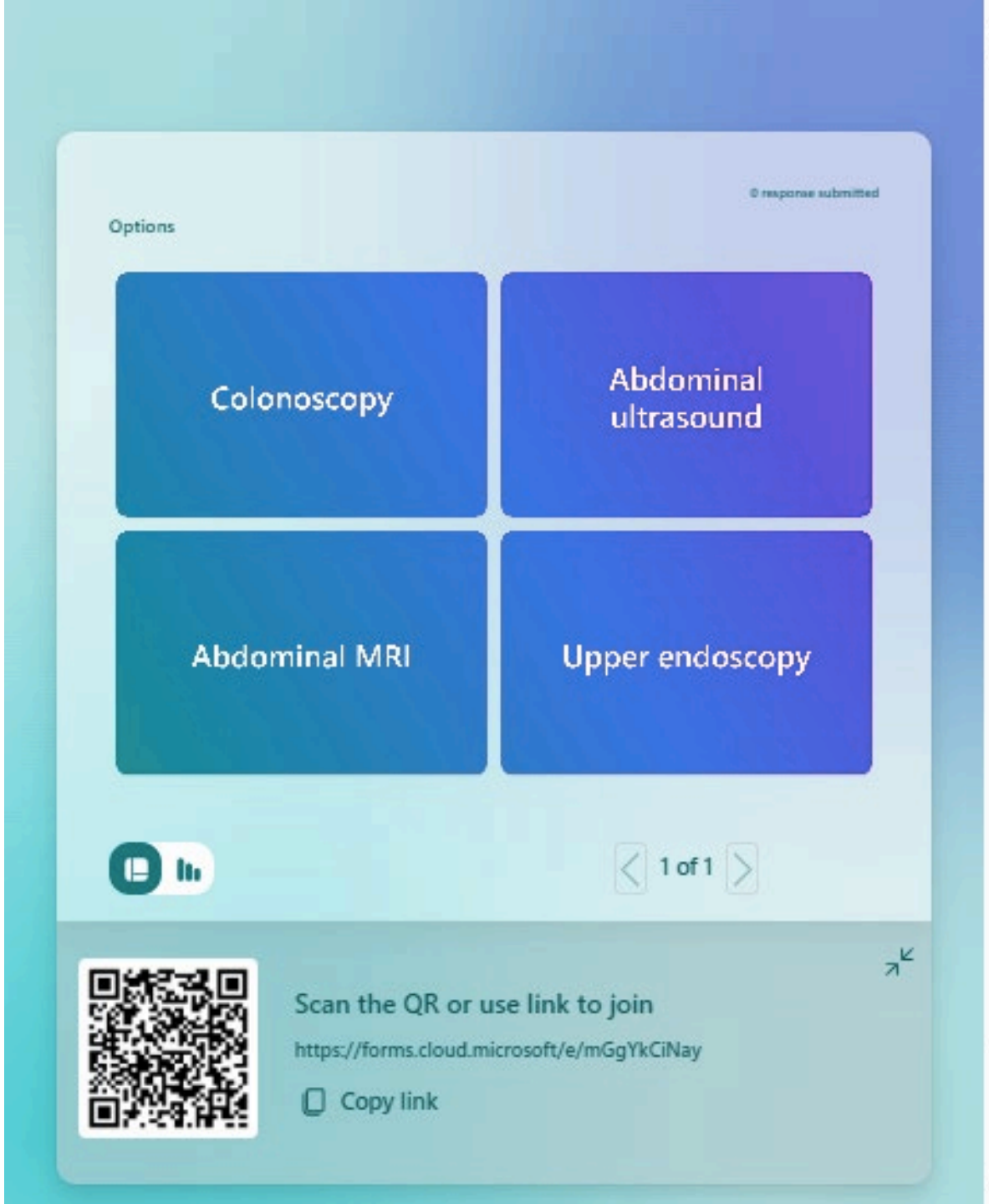


- Biopsies can be taken in case of gastric mucosal ulcers
- In patients with suspected perforation, the method of choice is CT (contrast is contraindicated!!)

Case 2

A 46-year-old woman presents with:

- Severe pain in the right upper abdomen
- Started 3 hours after a fatty meal
- Fever (38.7°C)
- Nausea and vomiting
- Positive Murphy's sign
- Elevated CRP and leukocyte count



Acute cholecystitis

Patient's history

- Pain after fatty meals, nausea, vomiting, fever
- Physical finding: Murphy's sign
- Laboratory: signs of inflammation

Abdominal US



The image is suggestive of acute cholecystitis:

- Thickening of the gallbladder wall (5 mm or greater)
- Accumulation of pericholecystic free fluid

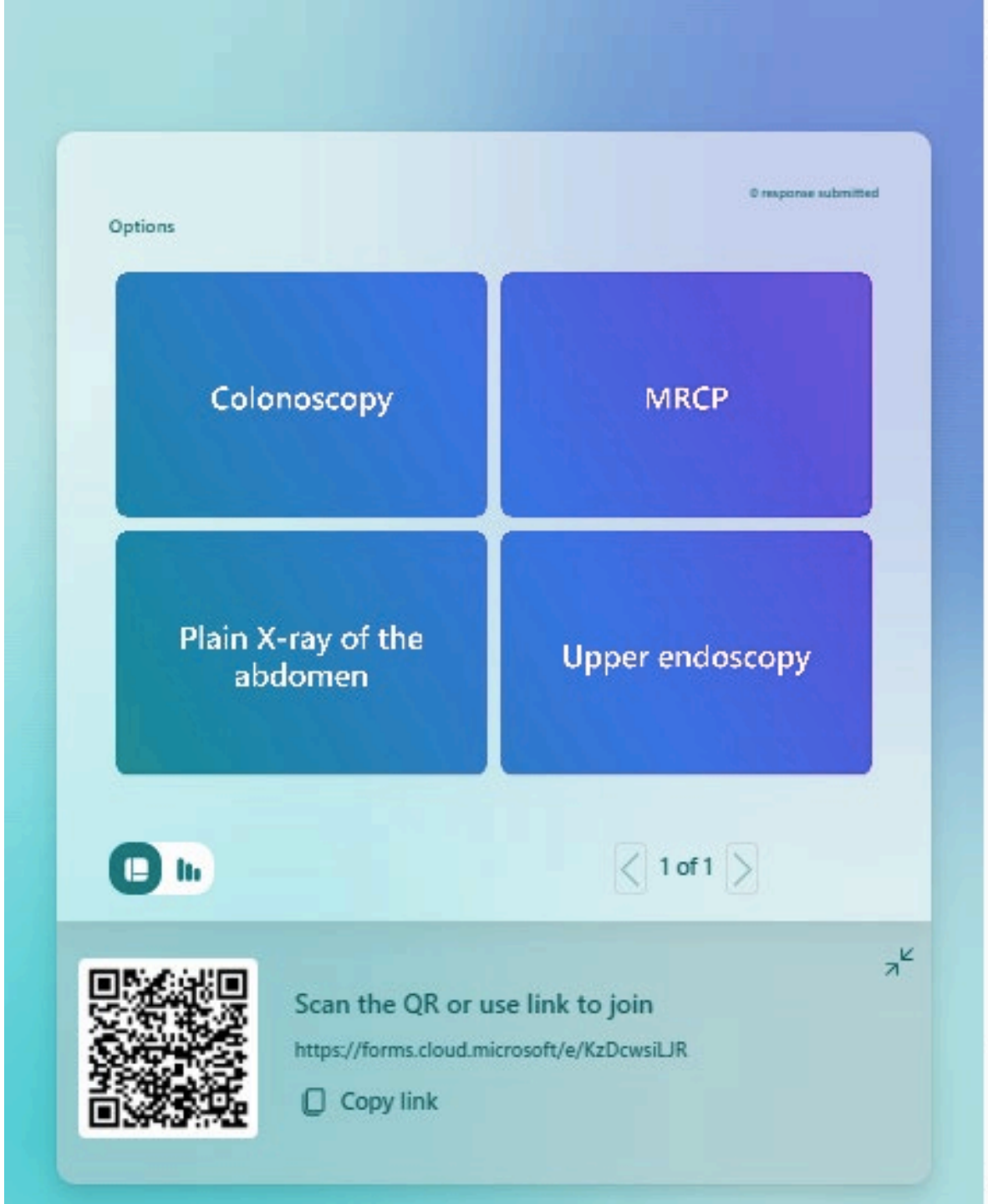
Case 3

55-year-old woman presents with:

- Right upper quadrant pain
- Jaundice
- Dark urine
- Pale stools

Laboratory findings:

- Bilirubin ↑
- ALP ↑
- GGT ↑



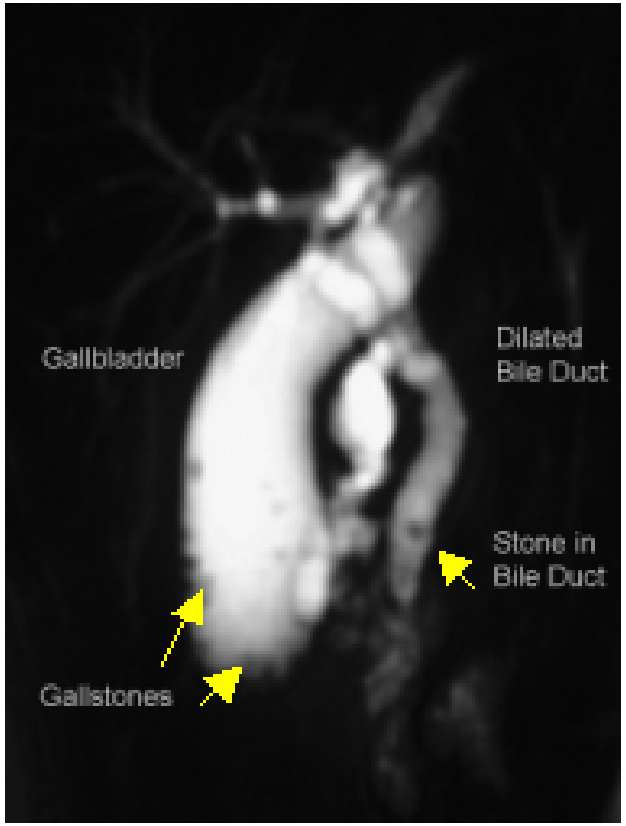
Choledocholithiasis

Patient's history

- Right upper quadrant pain and jaundice
- Laboratory: Elevated levels of bilirubin and particularly ALP & GGT

ERCP is being abandoned as it is invasive

Magnetic resonance cholangiopancreatography (MRCP)



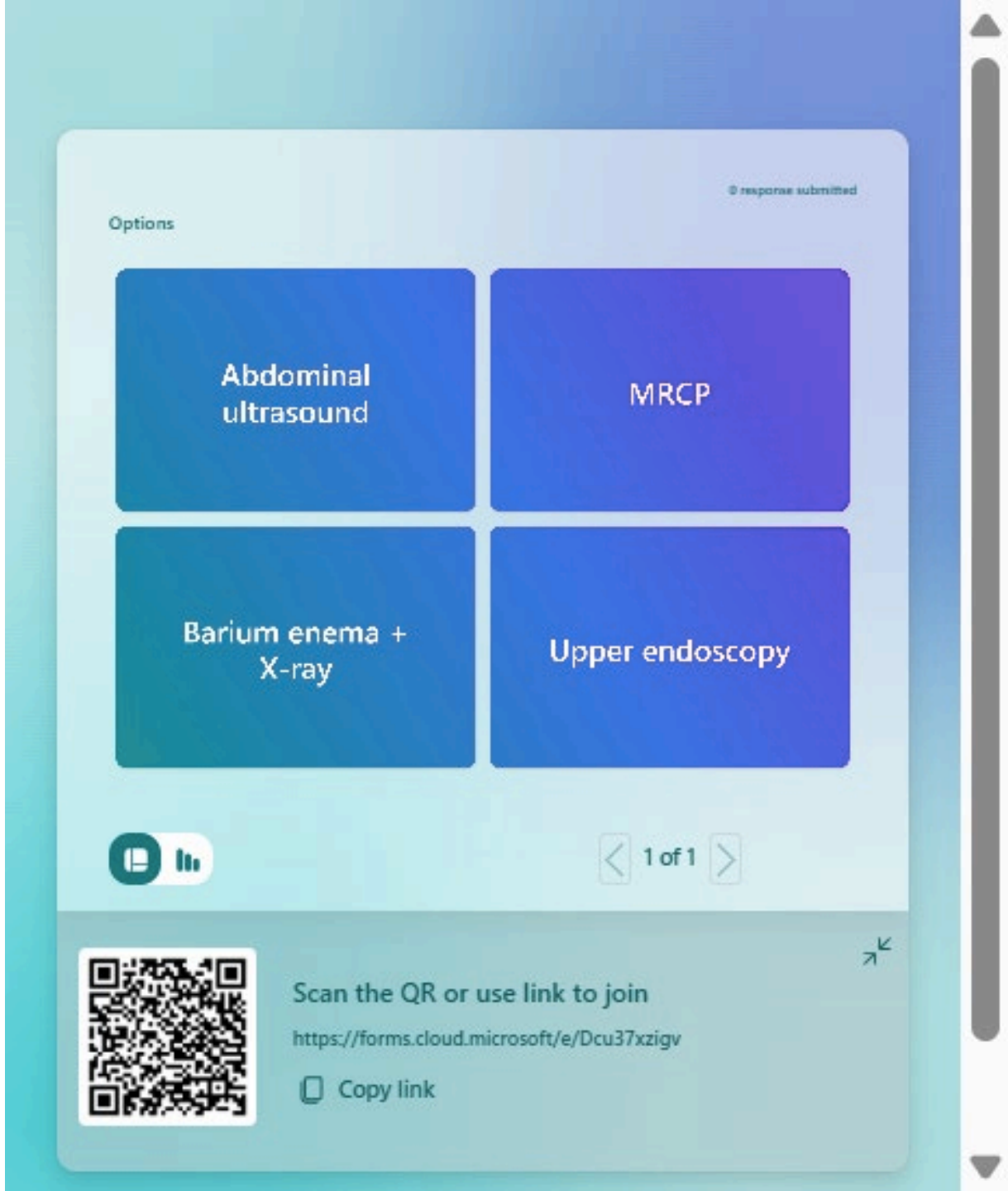
Case 4

A 61-year-old man presents with:

- Long history of alcohol misuse
- Fatigue
- Mild jaundice

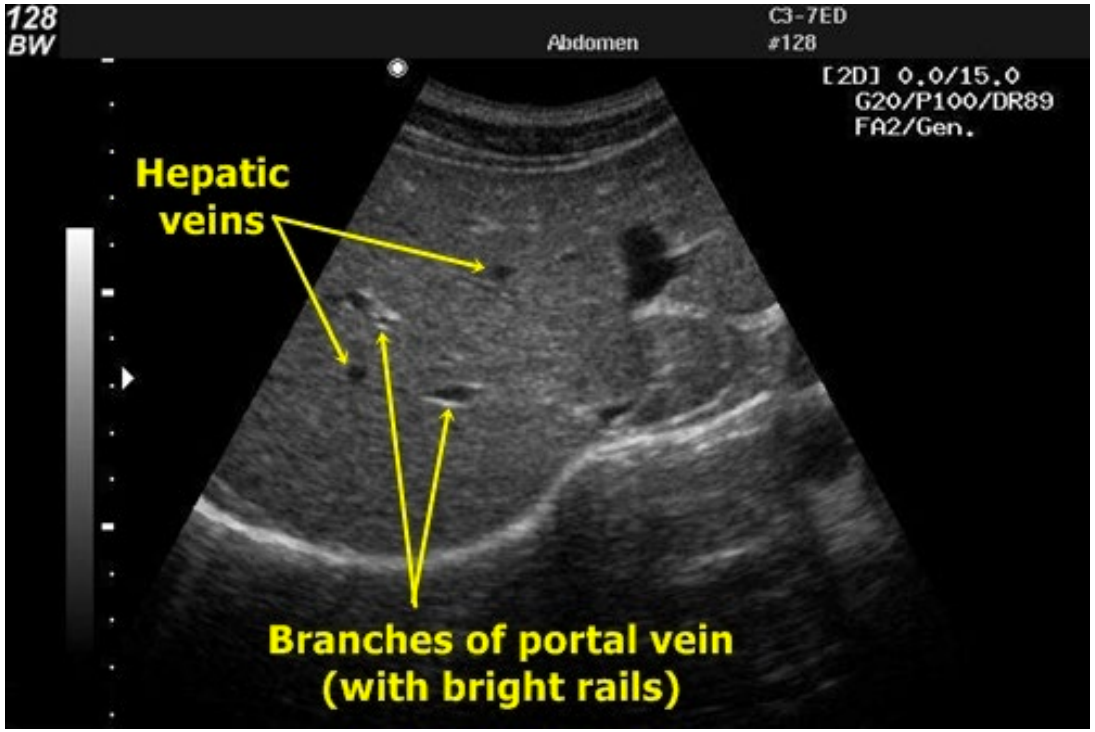
Physical examination:

- Ascites
- Enlarged spleen

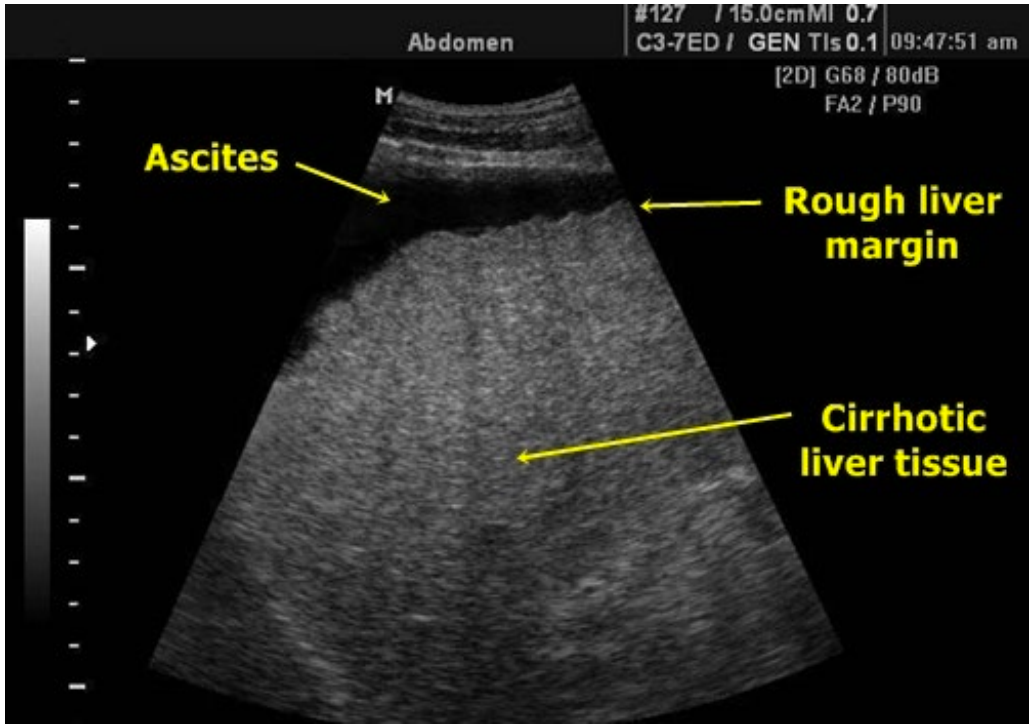


Liver cirrhosis

Abdominal US



Normal liver

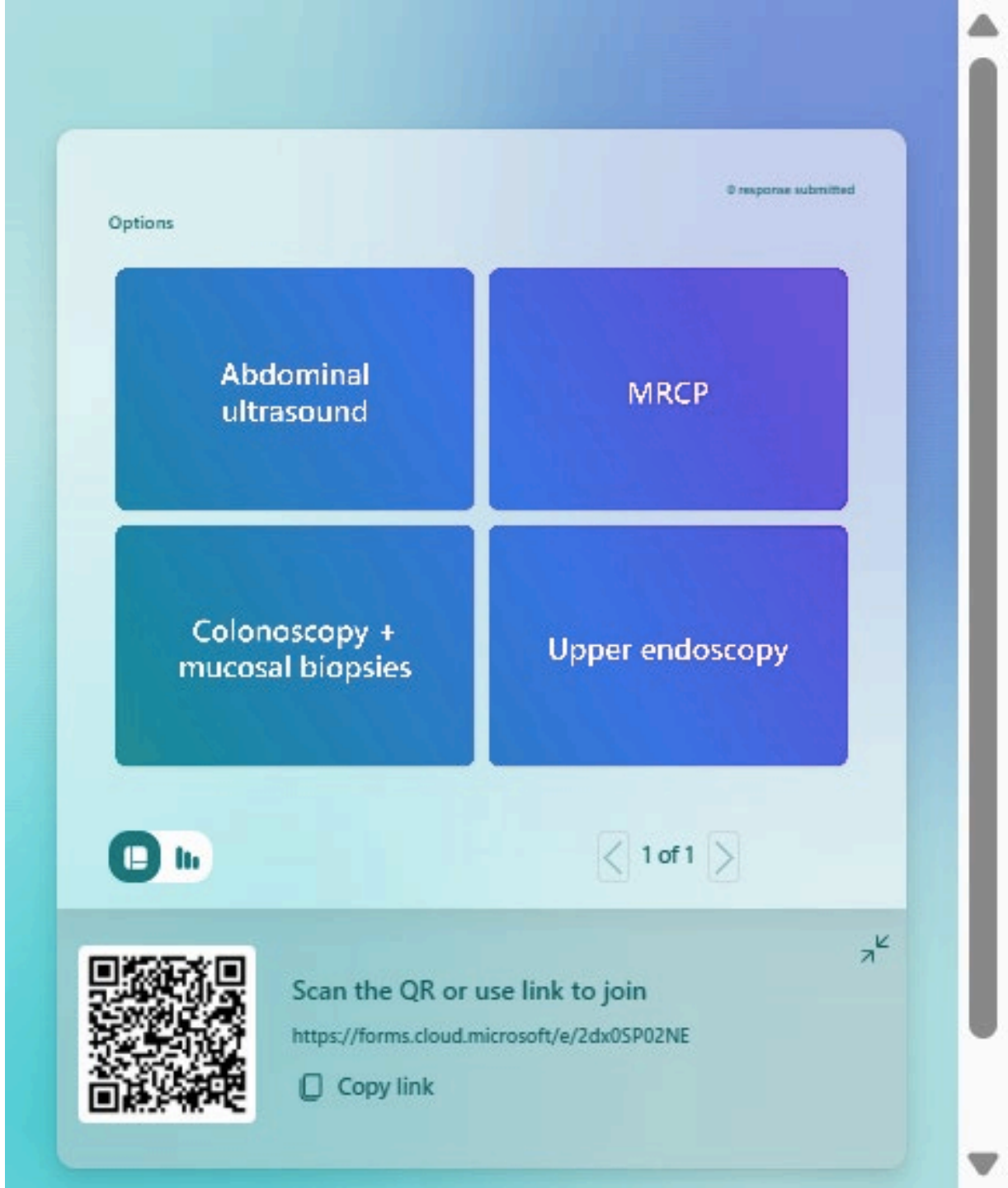


Cirrhotic liver

Case 5

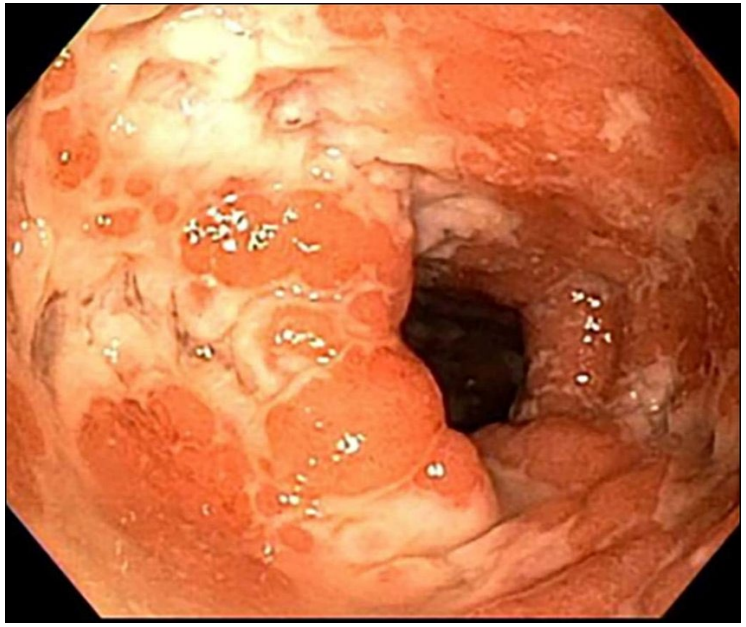
A 24-year-old woman presents with:

- Chronic diarrhea (÷blood)
- Weight loss
- Recurrent abdominal pain
- Elevated F-calprotectin



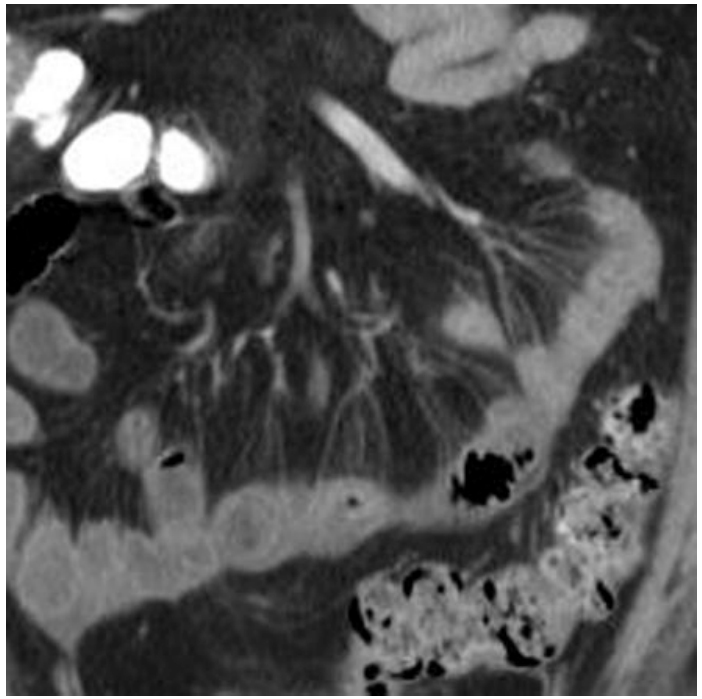
IBD (Crohn's disease)

Colonoscopy (+ biopsies)



Findings:
Skip lesions (transmural), reduction of lumen

Follow up: MRI /CT

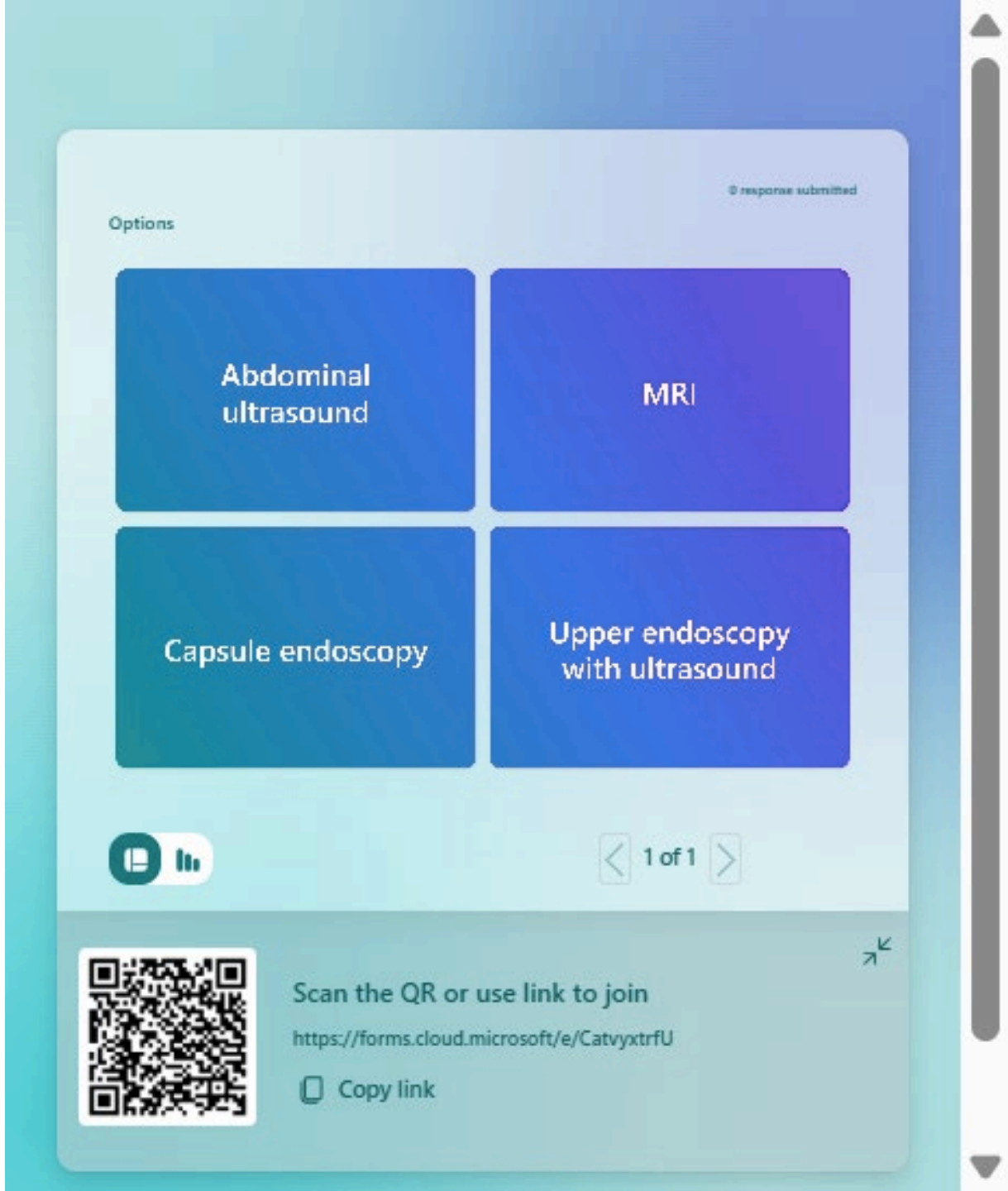


Hypervascular appearance of the mesentery in active Crohn disease ("comb sign").
Fibrofatty proliferation and perivascular infiltration

Case 6

A patient has biopsy-confirmed rectal cancer.

The surgeon wants to know how deeply the tumor has invaded and whether lymph nodes are involved.



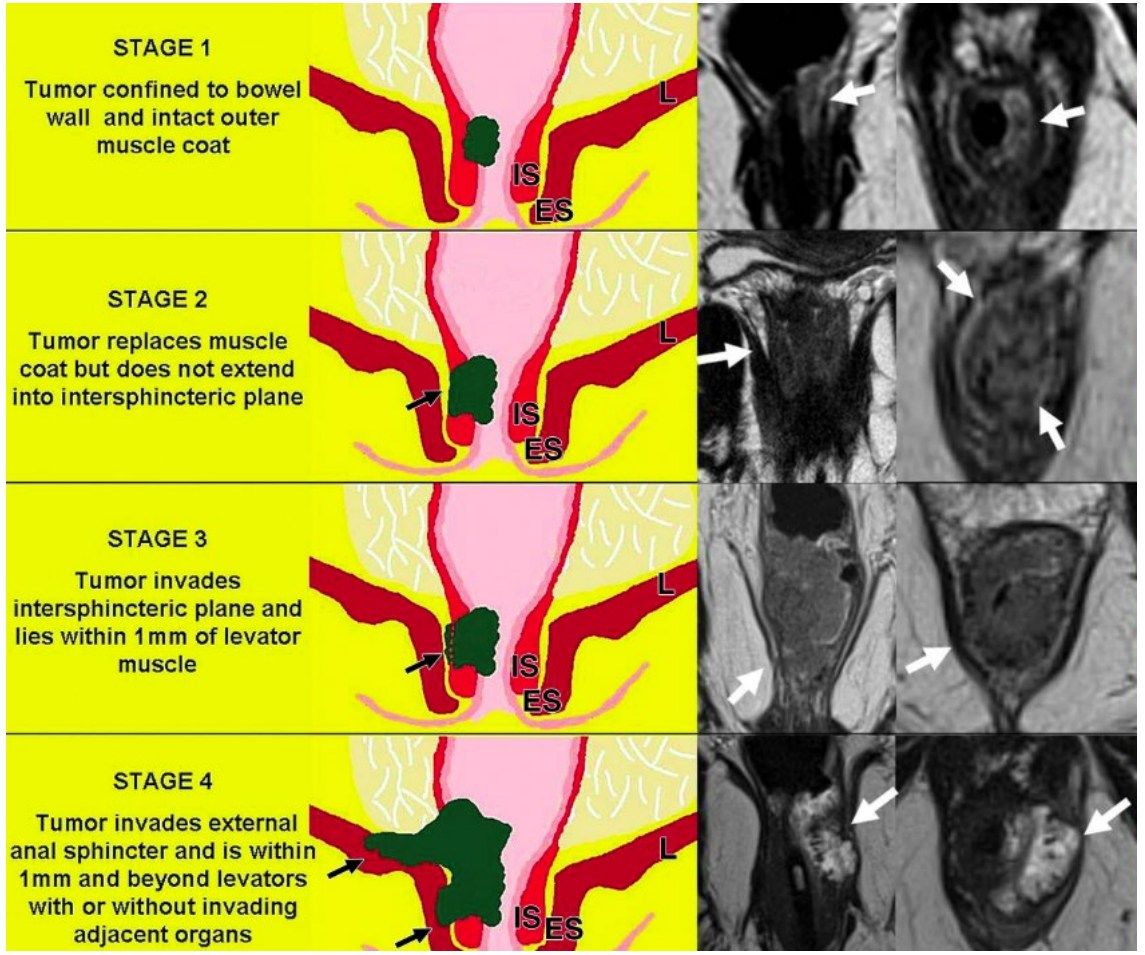
Tumor Staging

MRI

(it has the highest sensitivity)

Other options

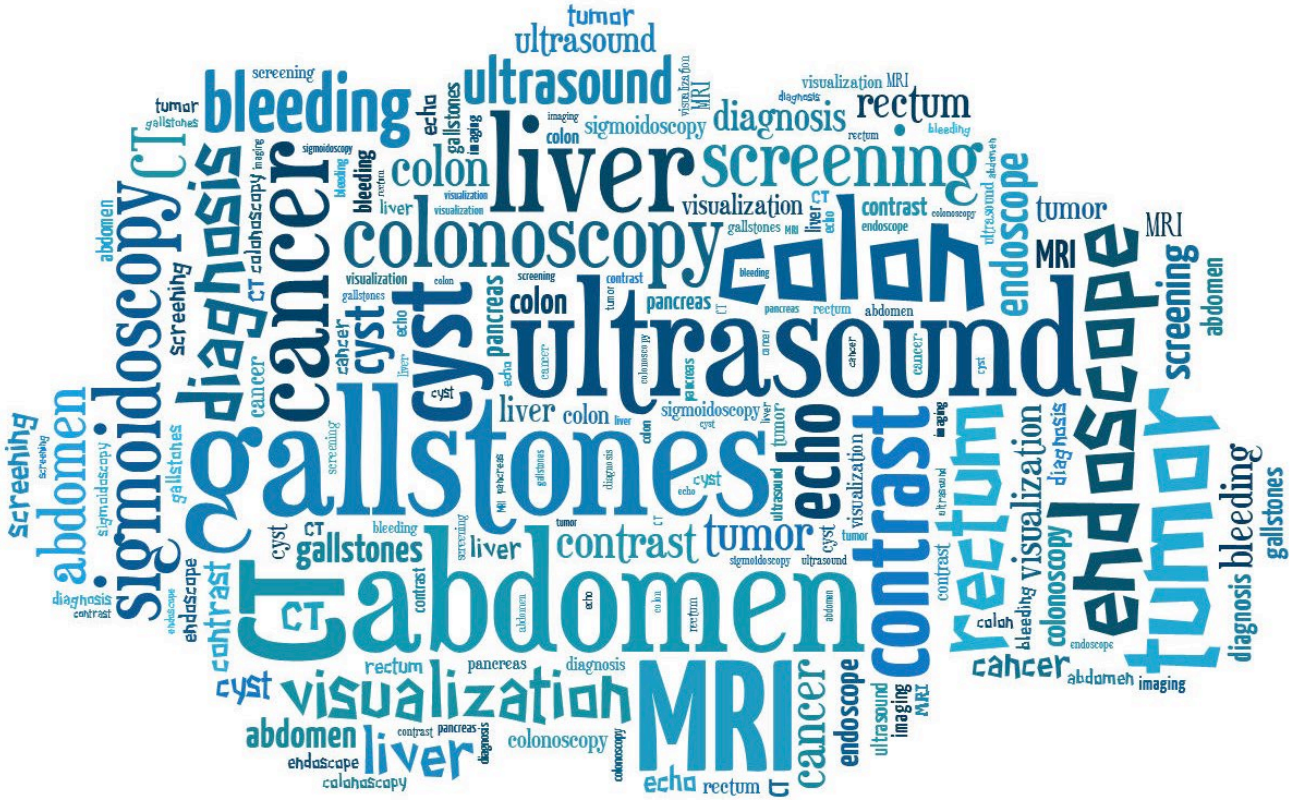
- CT (second line of choice)
- Endoscopic ultrasound

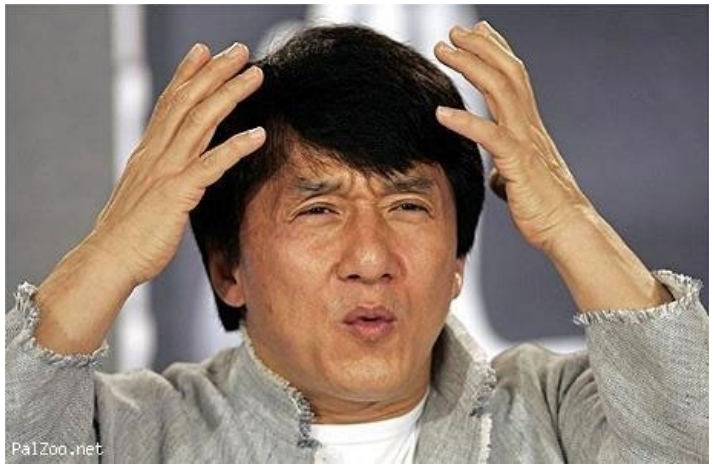


https://pubs.rsna.org/doi/10.1148/radiol.13121361

Factors that play a role in the selection

- Method sensitivity
- What is available?
- Patient tolerance (age, BMI, mental status, presence of implants, etc.)
- Expenses (cost)
- Risks of potential complications


tumor
ultrasound
bleeding
screening
ultrasound
echo
gallstones
liver
screening
sigmoidoscopy
diagnosis
rectum
colon
colonoscopy
visualization
contrast
endoscope
tumor
MRI
cancer
cyst
pancreas
colon
ultrasound
gallstones
abdomen
CT
liver
colon
sigmoidoscopy
echo
contrast
rectum
visualization
diagnosis
endoscope
tumor
screening
abdomen
bleeding
gallstones
CT
abdomen
contrast
tumor
gallstones
cyst
visualization
MRI
cancer
endoscope
ultrasound
imaging
liver
abdomen
colonoscopy
echo
rectum
CT


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Questions?